Homework 3 (of 5)

Probability and Statistics - solution

Questions 2

3.22 In order to control costs, a company wishes to study the amount of money its sales force spends entertaining clients. The following is a random sample of six entertainment expenses (dinner costs for four people) from expense reports submitted by members of the sales force. DS DinnerCost

$157 $132 $109 $145 $125 $139

a Calculate $\bar{x}$, s^2, and s for the expense data. In addition, show that the two different formulas for calculating s^2 give the same result.

b Assuming that the distribution of entertainment expenses is approximately normally distributed, calculate estimates of tolerance intervals containing 68.26 percent, 95.44 percent, and 99.73 percent of all entertainment expenses by the sales force.

c If a member of the sales force submits an entertainment expense (dinner cost for four) of $190, should this expense be considered unusually high (and possibly worthy of investigation by the company)? Explain your answer.

d Compute and interpret the z-score for each of the six entertainment expenses.

Questions 2

$$\mu = \frac{157 + 132 + 109 + 145 + 125 + 139}{6} = 134.5$$

$$\sigma^2 = \frac{(157 - 134.5)^2 + (132 - 134.5)^2 + (109 - 134.5)^2 + (145 - 134.5)^2 + (125 - 134.5)^2 + (139 - 134.5)^2}{6 - 1} = 276.7$$

$$\sigma = \sqrt{\sigma^2} = 16.64$$

For 68.26%: $(\mu - \sigma, \mu + \sigma) = (117.9,151.1)$

For 95.44%: $(\mu - 2\sigma, \mu + 2\sigma) = (101.2,167.8)$

For 99.73%: $(\mu - 3\sigma, \mu + 3\sigma) = (84.6,184.4)$

190 is unusually high. Because it is outside of the 99.73% tolerance region. The probability for spending outside of $= (84.6,184.4)$ is only 0.27% but it happens.

Use $z = \frac{x-\mu}{\sigma}$

Exercises (if time permits)

6.11 Suppose that an airline quotes a flight time of 2 hours, 10 minutes between two cities. Furthermore, suppose that historical flight records indicate that the actual flight time between the two cities, *x*, is uniformly distributed between 2 hours and 2 hours, 20 minutes. Letting the time unit be one minute.

(a) Write the formula for the probability curve of *x*.

(b) Graph the probability curve of *x*.

(c) Find $P(125 \leq x \leq 135)$.

(d) Find the probability that a randomly selected flight between the two cities will be at least five minutes late.

Answer

(a) $f(x) = \frac{1}{d-c} = \frac{1}{140-120} = 0.05$ for $120 \leq x \leq 140$.

(b) Graph the probability curve of *x*.

(c) $P(125 \leq x \leq 135) = \frac{135-125}{20} = \frac{10}{20} = 0.5$.

(d) $P(x \geq 130) = \frac{140-(130+5)}{140-120} = \frac{5}{20} = 0.25$.

Exercises (if time permits)

6.31 United Motors claims that one of its cars, the Starbird 300, gets city driving mileages that are normally distributed with a mean of 30 mpg and a standard deviation of 1 mpg. Let *x* denote the city driving mileage of a randomly selected Starbird 300.

(a)Assuming that United Motors' claim is correct, find $P(x \leq 27)$.

(b) If you purchase (randomly select) a Starbird 300 and your car gets 27 mpg in city driving, what do you think of United Motors' claim? Explain your answer.

Answer

(a) $P(x \leq 27) = P\left(z \leq \frac{27-30}{1}\right) = P(z \leq -3) = 0.00135$

(d) The claim is probably not true.

Exercises (if time permits)

6.33 A filling process is supposed to fill jars with 16 ounces of grape jelly. Specifications state that each jar must contain between 15.95 ounces and 16.05 ounces. A jar is selected from the process every half hour until a sample of 100 jars is obtained. When the fills of the jars are measured, it is found that $\bar{x} = 16.0024$ and $s = 0.02454$. Using $\bar{x}$ and s as point estimates of μ and σ, estimate the probability that a randomly selected jar will have a fill, x, that is out of specification. Assume that the process is in control and that the population of all jar fills is normally distributed.

Answer

(a) $P(x \leq 15.95) + P(x \geq 16.05) = P(x \leq 15.95) + 1 - P(x \leq 16.05)$

$$= P\left(z \leq \frac{15.95 - 16.0024}{0.02454}\right) + 1 - P\left(z \leq \frac{16.05 - 16.0024}{0.02454}\right)$$

$= P(z \leq -2.135) + 1 - P(z \leq 1.940) = 0.0164 + 1 - 0.9738 = 0.0426$